

MLOps Knowledge Pack (Condensed)

MLflow

- Tracking records experiments: parameters, metrics, artifacts, environment.
- Model Registry manages model lifecycle: versions, aliases/stages, approvals.
- Typical flow: Train -> Evaluate -> Register -> Validate -> Promote -> Deploy.
- Never deploy directly after training in production.

DVC

- Git versions code; DVC versions datasets and ML artifacts.
- Store only pointers in Git; data lives in remote storage (S3, GCS, Azure, MinIO).
- dvc.lock ensures reproducible pipelines.

KServe

- InferenceService is the main deployment resource.
- Supports canary traffic splitting and autoscaling.
- Models are typically loaded from object storage.

Kubernetes

- CI should build/test images and trigger Jobs, not run heavy training.
- Training should execute as Kubernetes Jobs or dedicated ML services.
- Use Deployments for inference workloads.

GitHub Actions

- Suitable for linting, testing, building containers, security scans, publishing images, triggering training workflows, and deployments.
- Avoid long-running training and large-scale data processing.

Monitoring

- Prometheus collects metrics.
- Grafana visualizes dashboards.
- Drift detection requires dedicated ML logic/tools in addition to infrastructure metrics.

Production Best Practices

- Compare challenger vs champion before promotion.
- Require validation gates.
- Canary deployments before full rollout.

- Rollback must be automated.
- Track lineage between code, data, model, and deployment.